AGENDA

- Diagonals / Upper - Lower Triangle in Matrix
- Rotate a matrix by 90 degrees
- Spirally traverse a matrix
- Make matrix beautiful
- Maximum square submatrix

Rotate a Matrix by 90 degrees

Given a square matrix **mat**[][] of size **n x n**. The task is to rotate it by **90 degrees** in an **anti-clockwise** direction without using any extra space.

Examples:

```
Input: mat[][] = [[0, 1, 2],
                  [3, 4, 5],
                  [6, 7, 8]]
Output: [[2, 5, 8],
         [1, 4, 7],
         [0, 3, 6]]
```

```
Input: mat[][] = [[1, 2],
                  [3, 4]]
Output: [[2, 4],
         [1, 3]]
```

Approach-1: Observe the pattern

1 2 3
4 5 6
7 8 9
becomes
3 6 9
2 5 8
1 4 7

Hence,
(0, 0) goes to (2, 0)
(0, 1) goes to (1, 0)
(0, 2) goes to (0, 0)

(1, 0) goes to (2, 1)
(1, 1) goes to (1, 1)
(1, 2) goes to (0, 1)

Hence, (i, j) goes to (n - j - 1, i)

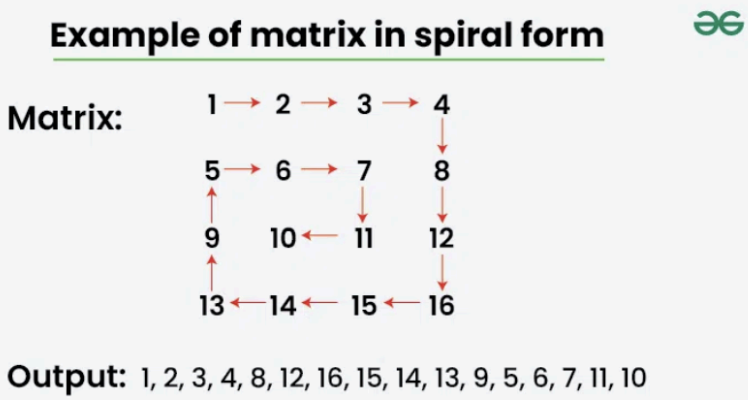
Approach-2: Correct the diagonals and then the rest

Spirally Traverse a Matrix

You are given a rectangular matrix **mat[][]** of size **n x m**, and your task is to return an array while **traversing** the matrix in **spiral** form.

Examples:

Input: mat[][] = [[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 14, 15, 16]]
Output: [1, 2, 3, 4, 8, 12, 16, 15, 14, 13, 9, 5, 6, 7, 11, 10]
Explanation:



Input: mat[][] = [[1, 2, 3, 4, 5, 6], [7, 8, 9, 10, 11, 12], [13, 14, 15, 16, 17, 18]]
Output: [1, 2, 3, 4, 5, 6, 12, 18, 17, 16, 15, 14, 13, 7, 8, 9, 10, 11]
Explanation: Applying same technique as shown above.

Make Matrix Beautiful

A **beautiful matrix** is defined as a square matrix in which the **sum** of elements in every row and every column is equal. Given a square matrix **mat[][]**, your task is to determine the **minimum number of operations** required to make the matrix beautiful.
In one operation, you are allowed to increment the value of **any single cell** by **1**.

Examples:

```
Input: mat[][] = [[1, 2],
                  [3, 4]]
Output: 4
Explanation:
Increment value of cell(0, 0) by 3,
Increment value of cell(0, 1) by 1.
Matrix after the operations: [[4, 3],
                              [3, 4]]
Here, sum of each row and column is 7.
Hence total 4 operation are required.
```

```
Input: mat[][] = [[1, 2, 3],
                  [4, 2, 3],
                  [3, 2, 1]]
Output: 6
Explanation:
Increment value of cell(0, 0) by 1,
Increment value of cell(0, 1) by 2,
Increment value of cell(2, 1) by 1,
Increment value of cell(2, 2) by 2.
Matrix after the operations: [[2, 4, 3],
                              [4, 2, 3],
                              [3, 3, 3]]
Here, sum of each row and column is 9.
Hence total 6 operation are required.
```

Maximum Square Submatrix

Problem Description

Given a 2D integer matrix **A** of size N x N find a B x B submatrix where B<= N and B>= 1, such that **sum of all the elements in submatrix is maximum**.

Problem Constraints

1 <= N <= 10^3.

1 <= B <= N

-10^2 <= A[i][j] <= 10^2.

Example Input

Input 1:

```
A = [
        [1, 1, 1, 1, 1]
        [2, 2, 2, 2, 2]
        [3, 8, 6, 7, 3]
        [4, 4, 4, 4, 4]
        [5, 5, 5, 5, 5]
    ]
B = 3
```

Input 2:

```
A = [
        [2, 2]
        [2, 2]
    ]
B = 2
```

Example Explanation

Explanation 1:

```
Maximum sum 3 x 3 matrix is
8 6 7
4 4 4
5 5 5
Sum = 48
```

Explanation 2:

```
Maximum sum 2 x 2 matrix is
 2 2
 2 2
 Sum = 8
```